Arithmetic in $\mathbb{Z}$

1. $-7 + 15$
2. $-24 - 18$
3. $32 - (-14)$
4. -9×7
5. $(-54) \div (-6)$
6. $-15 + 8 - (-12)$
7. $4 \times (-5) + 28 \div (-7)$
8. The morning temperature in a mountain town was $-6°$C. By afternoon, the temperature rose by $11°$C, and overnight it dropped by $9°$C. What was the final temperature?
9. A small business starts the week with a bank balance of $-£450$ (overdraft). On Tuesday, a deposit of $£1,200$ is made. On Thursday, an equipment bill of $£850$ is paid. What is the final bank balance at the end of the week?
10. Evaluate the expression: $-3 \times [(-8) - 4] + \dfrac{-36}{-9}$

Solutions

1. **8**
   Explanation: Adding a positive number to a negative number is equivalent to finding the difference between their absolute values and taking the sign of the larger number: $15 - 7 = 8$.

2. **-42**
   Explanation: Subtracting a positive integer from a negative integer (or adding two negative values) combines their magnitudes and retains the negative sign: $-24 - 18 = -(24 + 18) = -42$.

3. **46**
   Explanation: Subtracting a negative number is equivalent to adding its positive counterpart (the double negative rule): $32 - (-14) = 32 + 14 = 46$.

4. **-63**
   Explanation: The product of a negative integer and a positive integer is negative: $-9 \times 7 = -63$.

5. **9**
   Explanation: The quotient of two negative integers is positive: $(-54) \div (-6) = \dfrac{-54}{-6} = 9$.

6. **5**
   Explanation: Convert subtraction of a negative into addition first: $-15 + 8 - (-12) = -15 + 8 + 12$. Working left to right: $-15 + 8 = -7$, then $-7 + 12 = 5$.

7. **-24**
   Explanation: Following the order of operations, perform multiplications and divisions before addition:

   $$4 \times (-5) = -20$$
   $$28 \div (-7) = -4$$

   Combining them: $-20 + (-4) = -24$.

8. $-4°$**C**
   Explanation: Track the temperature changes step-by-step starting from -6:

   $$-6 + 11 = 5°\text{C} \quad \text{(afternoon rise)}$$
   $$5 - 9 = -4°\text{C} \quad \text{(overnight drop)}$$

   The final temperature is $-4°$C.

9. $-£100$ **(Overdraft of** $£100$**)**
   Explanation: Track the account balance chronologically:

   $$\begin{aligned} \text{Initial:} \quad & -450 \\ \text{After deposit:} \quad & -450 + 1,200 = 750 \\ \text{After bill payment:} \quad & 750 - 850 = -100 \end{aligned}$$

   The final balance is $-£100$ (an overdraft of $£100$).

10. **40**

Explanation: Evaluate following the order of operations:

(a) Evaluate inside the brackets: $(-8) - 4 = -12$.

(b) Multiply: $-3 \times (-12) = 36$.

(c) Evaluate the fraction division: $\dfrac{-36}{-9} = 4$.

(d) Add the results: $36 + 4 = 40$.